Perform Authorization using Grant and Revoke.

Name of Expt. 6	Perform Authorization using Grant and Revoke.

DCL- Data Control Language (DCL) consists of various commands which are related to data sharing and security of data in database.
DCL defines two commands,
Grant : Gives user access privileges to database.
Revoke : Take back permissions from user.

1. create a user and grant all permissions to the user.

a. create a user

mysql> CREATE USER 'username'@'localhost' IDENTIFIED BY 'password';
where:

- user – the name of the MySQL user which will be created
- password – the password which we want to assign to that user

b. grant all privileges
MYSQL> GRANT ALL PRIVILEGES ON table TO 'user'@'localhost';
Where
table – the name of the database table to which we grant access

- To display the privileged assigned to user, use `SHOW GRANTS` statement.
  Ex- SHOW GRANTS FOR super@localhost;

mysql>GRANT ALL PRIVILEGES ON *.* TO 'user'@'localhost';
user is granted to all databases.

c. To grant all privileges to the super@localhost user with grant option
Syntax- GRANT ALL ON *.* TO 'super'@'localhost' WITH GRANT OPTION;
The ON *.* clause means all databases and all objects in the databases. The WITH GRANT OPTION allows super@localhost to grant privileges to other users.
d. grant some privileges
mysql> GRANT SELECT, UPDATE, INSERT, DELETE ON <TABLE NAME> TO 'user'@'localhost';
e. List of MySQL Users
syntax – mysql> select user from mysql.user;

e. Remove an existing MySQL user
mysql>DROP USER 'user'@'localhost'

2. Revoke Privileges on Table
The REVOKE statement enables system administrators to revoke privileges from MySQL accounts.

a. revoke all permissions (except GRANT OPTION) on a table for a user
   mysql> REVOKE ALL ON <TABLE NAME> FROM 'user'@'localhost';

b. REVOKE DELETE, UPDATE ON <TABLE NAME> FROM 'user'@'localhost';
Ex- 1. mysql>REVOKE DELETE, UPDATE ON student FROM 'smita'@'localhost';

Implement GRANT and REVOKE statements in mysql.

Step 1 – CREATING USERS

To create another MySQL user by create user command other than 'root' user by following command steps:

1. At the command line, log in to MySQL as the root user:

```
mysql -u root -p
```

2. Type the MySQL root password, and then press Enter.
3. To create a database user, type the following command. Replace username with the user you want to create, and replace password with the user's password:

Syntax-

```
mysql>CREATE USER 'username'@'localhost' IDENTIFIED BY 'password';
example :
mysql> CREATE USER 'smita'@'localhost' IDENTIFIED BY '123'
Query OK, 0 rows affected (0.00 sec)
[Note : still you are logged in as root ]
```

Step 2 :

```
now select database for that you want to GRANT and REVOKE privileges by command
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| drupal78           |
| mysql              |
| college            |
| performance_schema |
+--------------------+
5 rows in set (0.04 sec)
```

then select that particular database that you want to GRANT and REVOKE privileges by command

```
mysql>USE college;
Database changed
mysql> show tables;
+------------------+
| Tables_in_college |
+------------------+
| dept             |
| insructor        |
| inst             |
```

| inst1 |
|+----------+
4 rows in set (0.00 sec)
Step 3 :
now GRANT and REVOKE privileges for the user that you want by *GRANT* and *REVOKE*
mysql> grant select on inst to 'smita'@'localhost';
Query OK, 0 rows affected (0.00 sec)

REVOKE
mysql> revoke delete on inst from 'smita'@'localhost';
Query OK, 0 rows affected (0.00 sec)

Step 4- Login to smita from another terminal
fcrit@fcrit-HP-15-Notebook-PC:~$ mysql -u smita -p
Enter password: 123

In mysql command line client need to follow following steps
Now use the above syntax discussed at the beginning. Give the username ‘smita’ and password ‘123’ which we created above to login. The command is as follows −

```
Microsoft Windows [Version 10.0.18362.959]
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C:\Users\SONY>cd\

C:\>cd program files\mysql\mysql server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -usmita -p
Enter password: ***
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.20 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

mysql> use college;

```
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> delete from inst where salary = 95000;
ERROR 1044 (42000): DELETE command denied to user 'smita'@'localhost' for table 'inst'.
mysql> select * from inst;
+----------+----------+
| name     | salary   |
+----------+----------+
| Einstein | 95000.00 |
| Wu       | 90000.00 |
| Katz     | 75000.00 |
| Crick    | 72000.00 |
+----------+----------+
4 rows in set (0.00 sec)
```

Conclusion	
Viva Questions	1) What are the commands under DCL